

WIND AND SOLAR POWER ENERGY SYSTEM

MODULE 1: PHYSICS OF WIND POWER

1.1 Historical Evolution of Wind Energy

Global Historical Perspective:

The utilization of wind energy represents one of humanity's oldest technological achievements. Ancient Persians (circa 500-900 AD) developed vertical-axis windmills with reed sails for grinding grain and pumping water. These designs spread through the Islamic world to China and later to Europe. European horizontal-axis windmills emerged around the 12th century, featuring mechanical innovations like fantails for automatic orientation and patent sails for variable speed control. The Industrial Revolution witnessed the American multi-bladed windpump's proliferation across farms, revolutionizing water abstraction for agriculture and livestock.

Indian Context:

India's wind energy journey commenced in the 1980s with demonstration projects in Gujarat and Maharashtra. The landmark 1994 policy framework established preferential tariffs, accelerated depreciation benefits, and wheeling/banking provisions, catalyzing commercial development. Tamil Nadu emerged as the pioneering state, harnessing consistent wind regimes in the Western Ghats. The state currently hosts approximately 40% of India's installed capacity. Subsequent growth phases saw Rajasthan, Gujarat, Karnataka, and Maharashtra developing substantial capacities.

Contemporary Statistics:

- **Global Capacity (2025):** ~1,200 GW cumulative installation
- **Leading Nations:** China (380+ GW), USA (140+ GW), Germany (65+ GW), India (45+ GW)
- **Indian Trajectory:** Current 45 GW against target of 60 GW by 2022 (achieved ~75%), with 140 GW target for 2030
- **Capacity Factor Variance:** 20-35% in India (site-dependent), compared to 35-45% in premium global sites

1.2 Fundamental Wind Physics

Kinetic Energy Conversion Principle:

Wind energy extraction relies on momentum exchange between moving air masses and rotor blades. The fundamental power equation derives from classical fluid dynamics:

$$P_{\text{wind}} = \frac{1}{2} \rho A v^3$$

Where:

ρ = air density (1.225 kg/m³ at sea level, 15°C)

A = swept area (πR^2 for horizontal-axis turbines)

V = undisturbed upstream wind velocity

Betz Limit Derivation and Implications:

German physicist Albert Betz (1919) established the theoretical maximum efficiency limit using actuator disk theory. Key assumptions include:

Homogeneous, incompressible, steady-state fluid flow

No frictional losses

Infinite number of blades

Uniform thrust distribution

Derivation Outline:

1. Mass flow rate through rotor: $\dot{m} = \rho A V_r$
2. Power extracted: $P = \frac{1}{2} \dot{m} (V_1^2 - V_2^2)$
3. Axial thrust: $T = \dot{m} (V_1 - V_2)$
4. From Bernoulli's principle: $V_r = \frac{V_1 + V_2}{2}$
5. Define induction factor: $a = \frac{V_1 - V_r}{V_1}$
6. Power coefficient: $C_p = \frac{P}{\frac{1}{2} \rho A V_1^3} = 4a(1 - a)^2$
7. Maximizing gives $a = \frac{1}{3}$ and $C_{p,max} = \frac{16}{27} \approx 0.593$

Practical Limitations:

Real turbines achieve 35-45% due to:

Profile drag losses (skin friction, pressure drag)

Tip losses (vortex shedding)

Wake rotation losses

Mechanical and electrical inefficiencies

Modern turbines typically operate at $C_p = 0.45-0.50$

1.3 Aerodynamic Design Parameters

Tip Speed Ratio (λ):

$$\lambda = (\omega R) / V$$

Where ω = rotational speed (rad/s), R = rotor radius, V = wind speed

TSR Classification:

High-speed rotors ($\lambda = 5-8$): 2-3 slender blades, electricity generation

Medium-speed rotors ($\lambda = 2-5$): 4-8 blades, mechanical drives

Low-speed rotors ($\lambda = 1-2$): 12-24 blades, water pumping

Solidity (σ): Ratio of total blade area to swept area

High solidity: High starting torque, low speed

Low solidity: Low starting torque, high efficiency at design speed

1.4 Control Methodologies

Stall Control (Passive):

Fixed-pitch blades with aerodynamic stall at high wind speeds

Mechanism: As wind speed increases, angle of attack increases until flow separation occurs, limiting power

Advantages: Mechanical simplicity, reliability, no pitch mechanism

Disadvantages: Imprecise power regulation, aerodynamic noise, reduced efficiency

Types:

Fixed stall: Blade angle optimized for single condition

Active stall: Minor pitch adjustments for better control

Pitch Control (Active):

Blades rotate along longitudinal axis to regulate angle of attack

Mechanism: Continuous adjustment maintains optimal angle across wind speeds

Control Strategies:

- Constant speed/variable pitch: Maintains synchronous speed
- Variable speed/variable pitch: Optimizes energy capture

- **Advantages:** Precise power regulation, better efficiency, reduced loads
- **Disadvantages:** Complex mechanism, higher cost, maintenance requirements

Yaw Control:

- Rotor orientation adjustment to face wind direction
- **Active yaw:** Electric or hydraulic drives with wind vane/sensor feedback
- **Passive yaw:** Tail vane or coning for small turbines

1.5 Wind Statistics and Probability Analysis

Weibull Distribution:

The two-parameter Weibull distribution accurately models wind speed frequency:

$$f(v) = \frac{k}{c} \left(\frac{v}{c} \right)^{k-1} e^{-(v/c)^k}$$

Shape parameter (k): 1.5-2.5 typically, indicates distribution peakedness

Scale parameter (c): Related to mean wind speed,

$$\bar{v} = c \Gamma(1 + 1/k)$$

Rayleigh Distribution:

Special case with k=2:

$$f(v) = \frac{2v}{c^2} e^{-(v/c)^2}$$

Often used when only mean wind speed is known.

Statistical Analysis Applications:

- **Site Assessment:** Energy yield estimation
- **Turbine Selection:** Matching turbine characteristics to wind regime
- **Reliability Analysis:** Extreme wind speed prediction for structural design
- **Grid Planning:** Intermittency and variability quantification

Cumulative Distribution Functions:

Probability that wind speed $\leq v$

$$F(v) = 1 - e^{-(v/c)^k}$$

Power duration curve: Integration of power curve with wind distribution

Capacity factor calculation: $CF = (\text{Actual energy}) / (\text{Rated power} \times 8760)$

:IMPORTANT QUESTION:

1.1 Historical Development and Current Status

Q1: What is the historical significance of wind energy development in India?

A1: India's wind energy journey began in the 1980s with demonstration projects in Gujarat and Maharashtra. The landmark 1994 policy framework established preferential tariffs, accelerated depreciation benefits, and wheeling/banking provisions, catalyzing commercial development. Tamil Nadu emerged as the pioneering state, harnessing consistent wind regimes in the Western Ghats, currently hosting approximately 40% of India's installed capacity. Subsequent growth phases saw Rajasthan, Gujarat, Karnataka, and Maharashtra developing substantial capacities, achieving 45 GW against a target of 140 GW by 2030.

Q2: Explain the Betz limit with its derivation assumptions.

A2: The Betz limit (59.3%) is the theoretical maximum efficiency for wind energy conversion, derived by Albert Betz in 1919 using actuator disk theory.

Assumptions:

1. Homogeneous, incompressible, steady-state fluid flow
2. No frictional losses
3. Infinite number of blades
4. Uniform thrust distribution
5. One-dimensional flow

The derivation shows that maximum power extraction occurs when downstream velocity is one-third of upstream velocity, yielding $C_{p,max} = 16/27 \approx 0.593$

Q3: Differentiate between stall and pitch control mechanisms.

A3:

Parameter	Stall Control	Pitch Control
Mechanism	Fixed blades, aerodynamic stall at high speeds	Adjustable blade angle
Complexity	Simple, passive	Complex, active
Efficiency	Lower at varying winds	Higher, maintains optimal angle
Cost	Lower initial cost	Higher due to mechanisms
Maintenance	Minimal	Regular maintenance needed
Power Regulation	Imprecise	Precise

Q4: Describe Weibull distribution parameters and their significance.

A4: The Weibull distribution

$$f(v) = \frac{k}{c} \left(\frac{v}{c}\right)^{k-1} e^{-(v/c)^k} \quad \text{has:}$$

Shape parameter (k): 1.5-2.5 typically, indicates distribution peakedness

Scale parameter (c): Related to mean wind speed $\bar{v} = cT$ **Significance:** Models wind speed frequency for energy yield estimation, turbine selection, extreme wind prediction, and reliability analysis.

NUMERICAL PROBLEMS:

Q1: A wind turbine with rotor diameter 80m operates in wind speed 12 m/s. Air density is 1.225 kg/m³. Calculate:

- a) Theoretical power available in wind
- b) Maximum power extractable (Betz limit)
- c) Actual power if power coefficient is 0.45

Solution:

a) Swept area: $A = \pi r^2 = \pi(40)^2 = 5026.55 \, m^2$

$$P_{wind} = \frac{1}{2} \rho A v^3 = 0.5 \times 1.225 \times 5026.55 \times (12)^3 = 5,328,000 \, W = 5.33 \, MW$$

b) $P_{Betz} = 0.593 \times 5.33 = 3.16 \, MW$

c) $P_{actual} = 0.45 \times 5.33 = 2.40 \, MW$

Q2: A 2 MW wind turbine produces 0.5 MW at wind speed 6 m/s. What power will it produce at 8 m/s assuming cubic relationship?

Solution:

$$\text{Power} \propto v^3$$

$$\frac{P_8}{P_6} = \left(\frac{8}{6}\right)^3 = 2.37$$

$$P_8 = 0.5 \times 2.37 = 1.185 \, MW$$

Q3: A site has Weibull parameters $k=2$ (Rayleigh) and $c=7$ m/s. Calculate:

- a) Probability of wind speed between 5-10 m/s**
- b) Most probable wind speed**
- c) Wind speed carrying maximum energy**

Solution:

$$a) P(5 < v < 10) = e^{-(5/7)^2} - e^{-(10/7)^2} = 0.489 - 0.041 = 0.448 \text{ (44.8\%)}$$

$$b) \text{ Most probable: } v_{mp} = c\sqrt{2} = 7 \times \sqrt{2} = 9.90 \text{ m/s}$$

$$c) \text{ Max energy: } v_{maxE} = c\sqrt[3]{3} = 7 \times \sqrt[3]{3} = 10.10 \text{ m/s}$$

Q4: A 3 MW wind turbine produces 7.5 GWh annually. Calculate its capacity factor.

Solution:

$$CF = \frac{\text{Annual Energy}}{\text{Rated Power} \times 8760} = \frac{7.5 \times 10^6}{3000 \times 8760} = 0.285 = 28.5\%$$

Q5: Calculate tip speed ratio for a turbine with radius 40m, rotational speed 15 rpm, and wind speed 10 m/s.

Solution:

$$\omega = (15 \times 2\pi) / 60 = 1.571 \text{ rad/s}$$

$$\lambda = (\omega R) / v = (1.571 \times 40) / 10 = 6.28$$

Q6: Compare power output at 1000m altitude ($\rho=1.112$ kg/m³) versus sea level ($\rho=1.225$ kg/m³) for same turbine and wind conditions.

Solution:

$$\frac{P_{1000}}{P_{sea}} = \frac{1.112}{1.225} = 0.908$$

9.2% decrease at altitude

Q7: A site with Rayleigh distribution (mean=7.5 m/s) has turbine with cut-in 4 m/s, rated 12 m/s, cut-out 25 m/s. Calculate operating time fraction.

Solution:

$$\text{For Rayleigh: } c = \frac{\bar{v}}{\sqrt{\pi/2}} = \frac{7.5}{1.253} = 5.986 \text{ m/s}$$

$$P(4 < v < 25) = e^{-(4/5.986)^2} - e^{-(25/5.986)^2} = 0.674 - 0.000 = 0.674$$

67.4% operating time

Q8: Compare wind power density between Site A ($v_{avg}=6$ m/s) and Site B ($v_{avg}=8$ m/s).

Solution:

$$\frac{P_B}{P_A} = \left(\frac{8}{6}\right)^3 = 2.37$$

Site B has 137% higher power density

Q9: Design a turbine rotor to produce 1.5 MW at 9 m/s with $C_p=0.48$. Calculate required diameter.

Solution:

$$P = \frac{1}{2} \rho A v^3 C_p$$

$$1.5 \times 10^6 = 0.5 \times 1.225 \times A \times 9^3 \times 0.48$$

$$A = 698.2 \text{ m}^2$$

$$D = 2\sqrt{A/\pi} = 2\sqrt{698.2/\pi} = 29.8 \text{ m}$$

Q10: A 2 MW turbine has capacity factor 32%. Calculate annual energy production and revenue at ₹4/kWh.

Solution:

$$E = 2000 \times 0.32 \times 8760 = 5,610,240 \text{ kWh} = 5.61 \text{ GWh}$$

$$\text{Revenue} = 5.61 \times 10^6 \times 4 = ₹2.24 \text{ crores}$$

MODULE 2: WIND GENERATOR TOPOLOGIES

2.1 Modern Wind Turbine Classification

Fixed-Speed Wind Turbines (Type 1):

Configuration: Squirrel cage induction generator directly grid-connected

Speed Operation: $\pm 1\%$ slip variation only

Power Control: Stall or active stall regulation

Reactive Compensation: Capacitor banks for power factor correction

Advantages: Simplicity, robustness, low cost, high reliability

Disadvantages: Limited speed range, poor power quality, high mechanical stress, limited grid support

Variable-Speed Wind Turbines:

Type 2: Limited Variable Speed

Wound rotor induction generator with variable rotor resistance

Speed variation: $\pm 10\%$ via power electronics in rotor circuit

Better grid compatibility than Type 1

Still requires reactive compensation

Type 3: Doubly-Fed Induction Generator (DFIG)

Architecture: Wound rotor with back-to-back partial scale converter (30% rating)

Speed Range: $\pm 30\%$ around synchronous speed

Power Flow:

Stator: Direct to grid (70% power)

Rotor: Bidirectional via converter (30% power)

Control Capabilities:

Decoupled active/reactive power control

Four-quadrant operation

Grid fault ride-through with crowbar protection

Advantages: Cost-effective (partial converter), efficient, excellent grid support

Disadvantages: Slip rings maintenance, converter protection complexity

Type 4: Full Power Converter Interface

Configurations:

PMSG with full-scale converter

SCIG with full-scale converter

Speed Range: 0-100% (theoretically)

Architecture: Generator → rectifier → DC link → inverter → grid

Advantages:

Complete grid decoupling

Superior power quality

Advanced grid support capabilities

Gearbox elimination possible (direct drive)

Disadvantages: Higher converter cost (100% rating), efficiency penalty

2.2 Generator Technologies

Induction Generators:

Operating Principle: Rotational speed > synchronous speed for generation

Magnetization: Requires reactive power from grid/capacitors

Types:

Squirrel cage: Robust, simple, fixed speed

Wound rotor: External resistance/converter access, variable speed

Doubly-Fed Induction Generator (DFIG) Detailed Analysis:

Rotor Circuit: Three-phase winding with slip rings

Converter System:

Rotor-side converter (RSC): Controls torque/speed

Grid-side converter (GSC): Controls DC link voltage and reactive power

Common DC bus with capacitors

Control Strategies:

Vector control with stator flux orientation

Maximum power point tracking (MPPT) through speed control

Grid synchronization using phase-locked loop (PLL)

Fault Protection:

Crowbar circuit: Shorts rotor during voltage dips

DC chopper: Limits overvoltage on DC bus

Demagnetization control: Rapid flux reduction

Permanent Magnet Synchronous Generator (PMSG):

Construction: Rotor with permanent magnets (NdFeB, SmCo)

Excitation: No external DC supply needed

Topologies:

Surface-mounted PM: Simple, non-salient

Interior PM: Robust, salient poles, higher torque density

Advantages:

High efficiency (no excitation losses)

High power density

Excellent low-speed performance

Direct drive capability

Disadvantages:

High cost of rare-earth magnets

Demagnetization risk at high temperatures

Complex manufacturing

Brushless Doubly-Fed Reluctance Machine (BDFRM):

Principle: Two stator windings with different pole numbers

Power winding: Grid-connected

Control winding: Converter-fed

Rotor: Simple reluctance structure (no windings/magnets)

Advantages: Brushless operation, rugged construction, fault tolerance

Applications: Medium-speed drives, pump storage

2.3 Power Electronics Converters

Converter Topologies:**Two-Level Voltage Source Converter:**

- Standard IGBT-based topology
- PWM switching (2-6 kHz)
- LC filters for harmonic mitigation

Three-Level Neutral Point Clamped (NPC):

- Reduced voltage stress per switch
- Better harmonic performance
- Higher efficiency for medium voltage

Matrix Converter:

- Direct AC-AC conversion
- No DC link components
- Bidirectional power flow
- Compact design

Modular Multilevel Converter (MMC):

Scalable to high voltages

Excellent waveform quality

High reliability (redundancy)

Control Architectures:**Generator-side Control:**

- MPPT algorithms
- Torque/speed regulation
- Field-oriented control (FOC)

Grid-side Control:

- DC link voltage regulation
- Reactive power control
- Grid synchronization
- Harmonic compensation

2.4 Artificial Intelligence in Wind Energy

Wind Forecasting:

Input Data: Historical wind data, meteorological parameters, numerical weather predictions

ML Techniques:

- ANN: Multi-layer perceptrons for pattern recognition
- SVM: Regression for time-series prediction
- LSTM: Sequence learning for temporal dependencies
- Ensemble methods: Random Forest, Gradient Boosting

Forecasting Horizons:

- Very short-term (5 min-1 hour): Turbine control
- Short-term (1-72 hours): Grid dispatch
- Medium-term (3-7 days): Maintenance planning
- Long-term (weeks-months): Resource assessment

Condition Monitoring and Predictive Maintenance:

Data Sources: SCADA, vibration sensors, oil analysis, thermal imaging

AI Applications:

- Anomaly detection: Unsupervised learning for fault identification
- Remaining useful life (RUL) prediction: Regression models
- Fault diagnosis: Classification algorithms (bearing, gearbox, blade faults)

Benefits: Reduced downtime, optimized maintenance schedules, extended component life

Optimal Control Applications:

Pitch Control Optimization: Reinforcement learning for adaptive control

Yaw Misalignment Correction: Computer vision + ML for wake steering

Farm Layout Optimization: Genetic algorithms for wake loss minimization

:IMPOTANT QUESTION:

Q1: Compare Type 1, 2, 3, and 4 wind turbine configurations.

A1:

Type	Generator	Converter	Speed	Advantages	Disadvantages
Type 1	SCIG	None	Fixed	Simple, robust	Poor power quality
Type 2	WRIG	Rotor resistance	Limited variable	Better than Type 1	Slip losses
Type 3	DFIG	Partial (30%)	Variable $\pm 30\%$	Cost-effective, good control	Slip rings, protection complex
Type 4	PMSG/SCIG	Full (100%)	Full variable	Best performance, no gearbox possible	Expensive converter

Q2: Explain DFIG working with power flow during super-synchronous operation.

A2: During super-synchronous operation (rotor speed > synchronous speed):

1. Stator delivers 70% power directly to grid
2. Rotor delivers 30% power through back-to-back converter
3. Rotor-side converter controls torque/speed
4. Grid-side converter controls DC link voltage and reactive power
5. Power flows from rotor to grid via converter

Q3: What are the advantages of PMSG over DFIG?

A3:

1. No gearbox (direct drive possible)
2. Higher efficiency (no excitation losses)
3. Better low-speed performance
4. Superior grid support capabilities
5. No slip rings maintenance
6. Higher power density

Q4: Describe FRT implementation in DFIG systems.

A4: Fault Ride-Through implementation involves:

1. **Crowbar Circuit:** Shorts rotor windings during severe dips
2. **DC Chopper:** Dissipates excess energy from DC link
3. **Demagnetization Control:** Rapid flux reduction
4. **Grid Support:** Reactive current injection during faults
5. **Control Strategy Change:** Switches to voltage support mode

NUMERICAL PROBLEMS:

Q1: A 3 MW DFIG operates with 0.7 slip in super-synchronous mode. Calculate rotor power and direction of power flow.

Solution:

$$P_{rotor} = s \times P_{stator} = 0.7 \times 3 = 2.1 \text{ MW}$$

Power flows FROM rotor TO grid (positive slip)

Q2: A 2.5 MW DFIG has partial converter rating of 30%. Calculate converter capacity. If rotor speed is 20% above synchronous speed, calculate rotor power.

Solution:

$$\text{Converter capacity} = 0.3 \times 2500 = 750 \text{ kW}$$

$$\text{Rotor power} = 0.20 \times 2500 = 500 \text{ kW (within converter rating)}$$

Q3: Calculate synchronous speed for 4-pole generator at 50 Hz and 6-pole generator at 60 Hz.

Solution:

$$4\text{-pole, } 50 \text{ Hz: } N_s = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

$$6\text{-pole, } 60 \text{ Hz: } N_s = \frac{120 \times 60}{6} = 1200 \text{ rpm}$$

Q4: A 4-pole DFIG has $\pm 30\%$ speed range around synchronous speed. Calculate minimum and maximum operating speeds.

Solution:

$$N_s = 1500 \text{ rpm}$$

$$N_{min} = 1500 \times 0.7 = 1050 \text{ rpm}$$

$$N_{max} = 1500 \times 1.3 = 1950 \text{ rpm}$$

Q5: Wind rotor rotates at 18 rpm. Generator (4-pole, 50 Hz) requires 1500 rpm. Calculate gear ratio. If generator is 6-pole, what gear ratio needed?

Solution:

$$\text{Gear ratio} = \frac{1500}{18} = 83.33$$

$$\text{For 6-pole: } N_s = 1000 \text{ rpm, Gear ratio} = \frac{1000}{18} = 55.56$$

Q6: A 100 MW wind farm operates at 0.95 power factor. Calculate reactive power. If PF improves to 0.98, how much reactive power reduced?

Solution:

$$\text{At } 0.95 \text{ PF: } \theta = \cos^{-1}(0.95) = 18.19^\circ, Q = 100 \times \tan(18.19^\circ) = 32.87 \text{ MVAr}$$

$$\text{At } 0.98 \text{ PF: } \theta = \cos^{-1}(0.98) = 11.48^\circ, Q = 100 \times \tan(11.48^\circ) = 20.29 \text{ MVAr}$$

$$\text{Reduction} = 12.58 \text{ MVAr}$$

Q7: A 3.3 kV line uses 3-level NPC converter. Calculate DC link voltage requirement.

Solution:

$$V_{dc} = \sqrt{2} \times V_{LL} = \sqrt{2} \times 3300 = 4667 \text{ V} = 4.67 \text{ kV}$$

Q8: PMSG produces 500 V line-line at 1000 rpm. What voltage at 750 rpm? What is the frequency at 750 rpm if 4-pole?

Solution:

$$\begin{aligned} \text{Voltage} \propto \text{speed: } V_{750} &= 500 \times \frac{750}{1000} = 375 \text{ V} \\ \text{Frequency} \propto \text{speed: } f &= \frac{P \times N}{120} = \frac{4 \times 750}{120} = 25 \text{ Hz} \end{aligned}$$

Q9: A 50 MW wind farm operates at 0.9 power factor. Calculate transformer MVA rating. If PF improves to 0.95, what transformer rating needed?

Solution:

$$\begin{aligned} \text{At 0.9 PF: } S &= \frac{P}{PF} = \frac{50}{0.9} = 55.56 \text{ MVA} \\ \text{At 0.95 PF: } S &= \frac{50}{0.95} = 52.63 \text{ MVA} \end{aligned}$$

Q10: A 2 MW turbine has 95% availability and 3% electrical losses. Calculate annual energy loss due to losses.

Solution:

$$\begin{aligned} \text{Annual energy without losses} &= 2000 \times 0.95 \times 8760 = 16,644,000 \text{ kWh} \\ \text{Loss energy} &= 16,644,000 \times 0.03 = 499,320 \text{ kWh} \end{aligned}$$

MODULE 3: THE SOLAR RESOURCE

3.1 Solar Radiation Fundamentals

Solar Constant and Extraterrestrial Radiation:

Solar Constant (G_0): $1367 \pm 21 \text{ W/m}^2$ (average at Earth's mean distance)

Variation: $\pm 3.4\%$ due to elliptical orbit (Earth-Sun distance: 147.1-152.1 million km)

Extraterrestrial Radiation:

$$G_{on} = G_{sc} \left[1 + 0.033 \cos \left(\frac{360n}{365} \right) \right]$$

Terrestrial Solar Radiation:

Atmospheric Effects:

Scattering: Rayleigh (molecules, blue sky), Mie (aerosols, haze)

Absorption: Ozone (UV), water vapor (IR), CO₂, oxygen

Reflection: Clouds, ground surfaces (albedo)

Spectral Distribution:

AM0: Space spectrum, 1367 W/m^2

AM1: Zenith path, 1000 W/m^2 (theoretical)

AM1.5: 37° from zenith, 1000 W/m^2 (standard testing)

UV (<400 nm): 5%

Visible (400-700 nm): 43%

Near IR (700-2500 nm): 52%

Radiation Components:

Beam/Direct (G_b): Straight line from sun, cast shadows

Diffuse (G_d): Scattered radiation, no shadows

Global (G): $G = G_b + G_d$

Albedo/Reflected (G_r): Ground-reflected radiation

3.2 Solar Geometry

Celestial Coordinate System:

Celestial Sphere: Imaginary sphere with Earth at center

Key Reference Planes:

- Equatorial plane: Earth's equator extended
- Ecliptic plane: Earth's orbital plane (tilt 23.45°)
- Horizon plane: Observer's local horizontal

Earth Motion Parameters:**Rotation:** 24-hour period, axis tilt 23.45°**Revolution:** 365.25 days, elliptical orbit

Perihelion: January 3-4 (147.1 million km)

Aphelion: July 4-5 (152.1 million km)

Declination Angle (δ):

Angle between sun's rays and equatorial plane

$$\delta = 23.45 \sin \left[\frac{360}{365} (284 + n) \right]$$

Where n = day number (Jan 1 = 1)

Key Dates:**Equinoxes ($\delta = 0^\circ$):** March 21, September 23**Summer Solstice ($\delta = +23.45^\circ$):** June 21 (Northern)**Winter Solstice ($\delta = -23.45^\circ$):** December 22**Latitude (ϕ):** Angular position north/south of equator**Classification:**

Low: 0-30° (tropical)

Mid: 30-60° (temperate)

High: 60-90° (polar)

Hour Angle (ω):

Angular displacement of sun east/west of local meridian

Solar noon: $\omega = 0^\circ$ **Morning:** ω negative**Afternoon:** ω positive**Conversion:** $\omega = 15^\circ \times (\text{Solar Time} - 12)$ **Zenith Angle (θ_z):**

Angle between vertical and sun direction

$$\cos \theta_z = \sin \phi \sin \delta + \cos \phi \cos \delta \cos \omega$$

Solar Altitude (α):

$$\alpha = 90^\circ - \theta_z$$

Azimuth Angle (γ_s):

Compass direction of sun (0° =North, 90° =East, 180° =South, 270° =West)

Sunrise and Sunset:**Sunset hour angle:**

$$\omega_s = \cos^{-1}(-\tan \phi \tan \delta)$$

Day length:

$$N = \frac{2}{15} \omega_s \text{ (hours)}$$

3.3 Solar Time Relationships**Equation of Time (E):**

Difference between apparent solar time and mean solar time

$$E = 9.87 \sin 2B - 7.53 \cos B - 1.5 \sin B \text{ (minutes)}$$

Where ,

$$B = \frac{360}{365}(n - 81)$$

Solar Time Calculation:

$$\text{Solar Time} = \text{Standard Time} + 4(L_{st} - L_{loc}) + E$$

Where:

L_{st} = standard meridian for time zone

L_{loc} = local longitude

Key Relationships:**Hour angle at sunrise/sunset:**

$$\omega_s = \cos^{-1}(-\tan \phi \tan \delta)$$

Maximum altitude at solar noon: $\alpha_{\max} = 90^\circ - |\phi - \delta|$

Optimum tilt angle: $\beta_{\text{opt}} = \phi \pm 15^\circ$ (seasonal adjustment)

3.4 Solar Radiation Estimation Methods

Empirical Models:

Angstrom-Prescott:

$$\frac{H}{H_0} = a + b \left(\frac{S}{S_0} \right)$$

Where S = sunshine hours, S_0 = day length

Hargreaves-Samani: Based on temperature difference

$$H = k_{RS} \cdot R_a \cdot \sqrt{T_{max} - T_{min}}$$

Bird Model: Spectral decomposition approach

Statistical Approaches:

Clearness index:

$$K_t = H/H_0$$

Diffuse fraction correlations

Probability distributions for daily radiation

Measurement Instruments:

Pyranometer: Global radiation (thermopile or photovoltaic)

Pyrheliometer: Direct beam (tracking required)

Sunshine Recorder: Campbell-Stokes (burning trace)

Spectroradiometer: Spectral distribution

Satellite-Based Estimation:

Geostationary satellites (Meteosat, GOES)

GIS mapping of solar resources

NASA SSE database (22-year global coverage)

:IMPORTANT QUESTION:

Q1: Explain solar constant and its variations.

A1: Solar constant is the extraterrestrial irradiance at Earth's mean distance = $1367 \pm 21 \text{ W/m}^2$.

Variations:

1. **Orbital eccentricity:** $\pm 3.4\%$ variation (147.1-152.1 million km)
2. **Solar activity:** $\pm 0.1\%$ over 11-year cycle
3. **Seasonal:** Calculated by

$$G_{on} = G_{sc} [1 + 0.033 \cos(360n/365)]$$

Q2: Define air mass and its significance in solar spectrum.

A2: Air Mass = path length through atmosphere relative to vertical path.

AM0: Space spectrum (1367 W/m²)

AM1: Vertical path (theoretical)

AM1.5: 37° from zenith, standard for testing (1000 W/m²)

Significance: Determines spectral distribution, with more scattering/absorption at higher AM.

Q3: Derive expression for solar day length.

A3: Sunset hour angle:

$$\omega_s = \cos^{-1}(-\tan \phi \tan \delta)$$

Day length: $N = 24 \cos \omega_s$ hours

Where ϕ = latitude, δ = declination angle

Q4: What are the components of terrestrial solar radiation?

A4:

Beam/Direct (G_b): Straight line from sun, cast shadows

Diffuse (G_d): Scattered radiation, no shadows

Global (G): $G = G_b + G_d$

Albedo/Reflected: Ground-reflected radiation

:NUMERICAL PROBLEMS:

Q1: Calculate declination angle on March 15 (n=74).

Solution:

$$\begin{aligned}\delta &= 23.45 \sin \left[\frac{360}{365} (284 + 74) \right] = 23.45 \sin \left[\frac{360}{365} \times 358 \right] \\ &= 23.45 \sin(353.1^\circ) = 23.45 \times (-0.103) = -2.42^\circ\end{aligned}$$

Q2: Calculate day length at latitude 28°N on June 21 (summer solstice).

Solution:

June 21: $\delta = 23.45^\circ$

$$\omega_s = \cos^{-1}(-\tan 28^\circ \tan 23.45^\circ) = \cos^{-1}(-0.249) = 104.3^\circ$$

$$N = \frac{2 \times 104.3}{15} = 13.91 \text{ hours}$$

Q3: At latitude 30°N, declination 20°, hour angle 45°, calculate solar altitude and zenith angle.

Solution:

$$\begin{aligned}\cos \theta_z &= \sin 30 \sin 20 + \cos 30 \cos 20 \cos 45 \\ &= (0.5 \times 0.342) + (0.866 \times 0.940 \times 0.707) = 0.171 + 0.576 = 0.747 \\ \theta_z &= \cos^{-1}(0.747) = 41.7^\circ \\ \alpha &= 90 - 41.7 = 48.3^\circ\end{aligned}$$

Q4: Location at 75°E longitude, time zone meridian 82.5°E. On March 15, equation of time is -10 minutes. Standard time is 10:30 AM. Find solar time.

Solution:

$$\begin{aligned}\text{Longitude correction} &= 4 \times (82.5 - 75) = 4 \times 7.5 = 30 \text{ minutes} \\ \text{Solar time} &= \text{Standard time} + \text{Longitude correction} + \text{Equation of time} \\ &= 10:30 + 00:30 + (-00:10) = 10:50 \text{ AM}\end{aligned}$$

Q5: Beam radiation is 800 W/m² at 45° zenith angle. Calculate beam radiation on horizontal surface.

Solution:

$$G_{bh} = G_b \times \cos \theta_z = 800 \times \cos 45^\circ = 800 \times 0.707 = 565.7 \text{ W/m}^2$$

Q6: In May, clearness index is 0.55. Extraterrestrial radiation is 40 MJ/m²/day. Calculate global radiation on horizontal surface.

Solution:

$$H = K_t \times H_0 = 0.55 \times 40 = 22 \text{ MJ/m}^2/\text{day}$$

Q7: Using Angstrom-Prescott model with a=0.25, b=0.50, day length 12 hours, sunshine duration 8 hours, calculate clearness index.

Solution:

$$K_t = a + b \left(\frac{S}{S_0} \right) = 0.25 + 0.50 \left(\frac{8}{12} \right) = 0.25 + 0.333 = 0.583$$

Q8: Find optimum tilt angle for fixed solar panel at latitude 20°N for maximum annual yield.

Solution:

$$\text{For annual maximum: } \beta_{opt} = \phi = 20^\circ$$

Q9: Earth is 3% closer to sun at perihelion. Calculate solar constant at perihelion.

Solution:

$$\begin{aligned} \text{Inverse square law: } \frac{G_{peri}}{G_{avg}} &= \left(\frac{1}{0.97} \right)^2 = 1.062 \\ G_{peri} &= 1367 \times 1.062 = 1452 \text{ W/m}^2 \end{aligned}$$

Q10: Global radiation 600 W/m², beam normal radiation 700 W/m², zenith angle 40°. Calculate diffuse radiation.

Solution:

$$\begin{aligned} \text{Beam on horizontal: } G_b &= 700 \times \cos 40^\circ = 700 \times 0.766 = 536.2 \text{ W/m}^2 \\ \text{Diffuse: } G_d &= G - G_b = 600 - 536.2 = 63.8 \text{ W/m}^2 \end{aligned}$$

MODULE 4: SOLAR PHOTOVOLTAIC SYSTEMS

4.1 PV Technology Classification

First Generation: Crystalline Silicon

Monocrystalline (c-Si):

- Czochralski process: Single crystal ingot growth
- Efficiency: 18-22% (laboratory: 26.7%)
- Characteristics: Uniform dark color, high purity
- Advantages: Highest efficiency, long lifespan (>25 years)
- Disadvantages: Highest cost, silicon waste

Polycrystalline (mc-Si):

- Casting process: Multi-crystal ingot
- Efficiency: 15-18%
- Characteristics: Speckled blue color, grain boundaries
- Advantages: Lower cost, better temperature coefficient
- Disadvantages: Lower efficiency, aesthetic limitations

Second Generation: Thin Films

Amorphous Silicon (a-Si):

- PECVD deposition on glass/metal/plastic
- Efficiency: 6-8% (initial), 4-6% (stable)
- Staebler-Wronski effect: Light-induced degradation (10-30%)
- Advantages: Low-temperature process, flexible substrates
- Applications: Consumer electronics, building integration

Cadmium Telluride (CdTe):

- Close-spaced sublimation or vapor transport
- Efficiency: 10-12% (module), 22.1% (cell record)
- Advantages: Lowest manufacturing cost, good spectral response
- Concerns: Cadmium toxicity, tellurium scarcity

Copper Indium Gallium Selenide (CIGS):

- Co-evaporation or sputtering
- Efficiency: 12-14% (module), 23.4% (cell record)
- Advantages: Highest thin-film efficiency, flexible
- Disadvantages: Complex stoichiometry control, indium scarcity

Third Generation: Emerging Technologies**Perovskite Solar Cells:**

- ABX_3 crystal structure (e.g., $CH_3NH_3PbI_3$)
- Efficiency progression: 3.8% (2009) → 25.7% (2024)
- Advantages: Solution processable, tunable bandgap
- Challenges: Stability, lead toxicity, scaling

Organic Photovoltaics (OPV):

- Polymer/fullerene blends
- Efficiency: 8-12%
- Advantages: Ultra-flexible, semitransparent, low-cost
- Disadvantages: Low efficiency, poor stability

Multi-junction Cells:

- III-V compounds (GaInP/GaAs/Ge)
- Efficiency: >40% (concentrated), 30-35% (non-concentrated)
- Applications: Space, concentrated PV (CPV)

Dye-Sensitized Solar Cells (DSSC):

- Photosensitized anode (TiO_2 + dye)
- Efficiency: 11-12%
- Advantages: Low-cost, works in diffuse light
- Disadvantages: Liquid electrolyte issues, stability

4.2 PV Cell Physics and Characteristics**Photovoltaic Effect:****Mechanism:**

- Photon absorption creates electron-hole pair
- Built-in electric field (p-n junction) separates charges
- External circuit allows current flow

Bandgap Energy (E_g):

- Si: 1.12 eV (optimal for solar spectrum)
- a-Si: 1.7 eV (adjustable by hydrogen content)
- CdTe: 1.45 eV
- CIGS: 1.0-1.7 eV (tunable by Ga/In ratio)

I-V Characteristics:

Equivalent Circuit: Single-diode model with series and shunt resistance

$$I = I_L - I_0 \left[\exp \left(\frac{V + IR_s}{nV_t} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}}$$

Where:

- I_L = Light-generated current
- I_0 = Diode saturation current
- R_s = Series resistance (contacts, bulk)
- R_{sh} = Shunt resistance (leakage)
- n = Ideality factor (1-2)
- $V_t = kT/q$ = Thermal voltage (26 mV at 300K)

Key Parameters:**Short Circuit Current (I_{sc}):**

Maximum current at V=0

Proportional to irradiance

Temperature coefficient: +0.04%/°C

Open Circuit Voltage (V_{oc}):

Maximum voltage at I=0

Logarithmic with irradiance

Temperature coefficient: -0.3%/°C

Maximum Power Point (MPP):

$$P_{max} = V_{mp} \times I_{mp}$$

Typically 70-80% of $V_{oc} \times I_{sc}$

Fill Factor (FF):

$$FF = \frac{V_{mp} I_{mp}}{V_{oc} I_{sc}}$$

Typical values: 0.7-0.8

Affected by R_s (reduces FF) and R_{sh} (increases FF)

Efficiency (η):

$$\eta = \frac{P_{max}}{P_{in}} = \frac{V_{oc} I_{sc} FF}{A \times G}$$

Temperature Effects:

Voltage: Decreases linearly ($\sim -0.33\%/^{\circ}\text{C}$ for c-Si)

Current: Slight increase ($\sim +0.04\%/^{\circ}\text{C}$)

Power: Net decrease (~ -0.4 to $-0.5\%/^{\circ}\text{C}$)

NOCT: Nominal Operating Cell Temperature (typically $45 \pm 2^{\circ}\text{C}$ at 800 W/m^2 , 20°C ambient, 1 m/s wind)

Irradiance Effects:

Current: Linear proportionality

Voltage: Logarithmic increase

Efficiency: Slight increase at low irradiance (better FF)

4.3 PV System Components and Configuration**Module Construction:****Cell → Module Process:**

Texturing (pyramids for light trapping)

Diffusion (p-n junction formation)

Anti-reflection coating (Si_3N_4 , MgF_2)

Metallization (screen printing)

Tabbing and stringing

Lamination (EVA encapsulation)

Framing and junction box

Array Configurations:

Series Connection: Increases voltage, current constant

Mismatch losses from shading/lowest current cell

Bypass diodes protect against hot spots

Parallel Connection: Increases current, voltage constant

Blocking diodes prevent reverse current

Lower mismatch sensitivity

Series-Parallel: Balance of voltage and current

Balance of System (BOS):

Mounting Structures:

Fixed tilt: Simple, low-cost

Single-axis tracking: E-W or N-S, 15-25% gain

Dual-axis tracking: Maximum yield, 30-40% gain, higher cost

Cabling: DC and AC wiring with proper sizing

Protection:

Fuses/circuit breakers

Surge protection devices (SPD)

Grounding and bonding

4.4 Power Electronics for PV Systems

DC-DC Converters:

Buck Converter: Step-down, $V_{out} < V_{in}$

Boost Converter: Step-up, $V_{out} > V_{in}$ (common for PV)

Buck-Boost Converter: Both step-up and step-down

Cuk Converter: Continuous input/output current

SEPIC: Single-ended primary inductance converter

Inverter Technologies:

Central Inverters:

Large systems (> 100 kW)

Single MPPT for entire array

Advantages: Low \$/W, high efficiency (98-99%)

Disadvantages: Single point failure, mismatch losses

String Inverters:

Per string MPPT (5-30 kW)

Advantages: Multiple MPPTs, better shade tolerance

Disadvantages: More components, higher installation cost

Microinverters:

Module-level power electronics (200-400 W)

Advantages: Maximum energy harvest, individual monitoring, safe DC voltage

Disadvantages: Higher cost, more components, reliability concerns

Power Optimizers:

DC-DC converter at module level

MPPT at module level, central inverter

Balance between microinverter and string inverter benefits

MPPT Algorithms:**Perturb and Observe (P&O):**

Periodically perturbs voltage and observes power change

Advantages: Simple implementation

Disadvantages: Oscillation around MPP, confusion during rapid irradiance changes

Incremental Conductance :

Based on $dP/dV = 0$ at MPP

Condition:

$$\frac{I}{V} + \frac{dI}{dV} = 0$$

Advantages: Accurate, no oscillation at steady state

Disadvantages: Complex computation, sensitive to noise

Fractional Methods:

FOCV: $V_{mp} = k_1 \times V_{oc}$ ($k_1 \approx 0.76-0.82$)

FSCC: $I_{mp} = k_2 \times I_{sc}$ ($k_2 \approx 0.92-0.95$)

Advantages: Simple, fast tracking

Disadvantages: Requires periodic measurement of V_{oc} or I_{sc}

Intelligent Methods:

Fuzzy logic, neural networks

Particle swarm optimization, genetic algorithms

Advantages: Good performance under partial shading

Disadvantages: Complex implementation, tuning required

4.5 AI Applications in PV Systems

Performance Monitoring and Fault Detection:

Thermal Imaging Analysis: Hot spot detection using CNN

I-V Curve Analysis: Machine learning for fault classification

Partial shading

Soiling

Degradation

Mismatch

Inverter faults

MPPT Optimization:

ANN-based MPPT: Training with historical data

Reinforcement Learning: Adaptive control under varying conditions

Ensemble Methods: Combining multiple algorithms

Yield Prediction and Forecasting:

Time-series Models: LSTM, GRU for short-term forecasting

Image-based Prediction: Sky cameras with CNN for cloud movement

Hybrid Models: Combining numerical weather predictions with ML

Optimal Cleaning Schedule:

Soiling Prediction: Regression models based on weather, location

Cost-benefit Analysis: Optimization of cleaning frequency

Robotic Cleaning: Path planning algorithms

:IMPORTANT QUESTION:

Q1: Compare crystalline silicon and thin-film PV technologies.

A1:

Parameter	c-Si	Thin-film
Efficiency	15-22%	6-14%
Cost	Higher	Lower
Temperature coeff.	-0.4 to -0.5%/°C	-0.2 to -0.3%/°C
Low-light performance	Poor	Better
Aesthetics	Opaque	Semi-transparent possible

Parameter	c-Si	Thin-film
Applications	Rooftop, utility	BIPV, flexible

Q2: Explain single-diode model of PV cell.

A2: Equivalent circuit includes:

1. Current source (I_L) for photocurrent
2. Diode for p-n junction
3. Series resistance (R_s) for contacts/bulk
4. Shunt resistance (R_{sh}) for leakage

Equation:

$$I = I_L - I_0 \left[\exp\left(\frac{V + IR_s}{nV_t}\right) - 1 \right] - \frac{V + IR_s}{R_{sh}}$$

Q3: What is NOCT and its significance?

A3: Nominal Operating Cell Temperature = Temperature reached by open-circuited cell under:

Irradiance: 800 W/m²

Ambient: 20°C

Wind: 1 m/s

Significance: Predicts operating temperature for performance calculation

Q4: Compare MPPT algorithms: P&O vs Incremental Conductance.

A4:

Aspect	P&O	IncCond
Principle	Perturb voltage, observe power	$dP/dV = 0$ at MPP
Accuracy	Medium	High
Oscillation	Yes at steady state	Minimal
Complexity	Low	Medium
Tracking speed	Medium	Fast
Irradiance change	Can lose track	Handles better

: NUMERICAL PROBLEMS:

Q1: A PV module has: $V_{oc}=40V$, $I_{sc}=8.5A$, $V_{mp}=32V$, $I_{mp}=7.8A$, area= 1.6 m^2 . Calculate fill factor and efficiency at STC (1000 W/m^2).

Solution:

$$FF = \frac{V_{mp}I_{mp}}{V_{oc}I_{sc}} = \frac{32 \times 7.8}{40 \times 8.5} = \frac{249.6}{340} = 0.734$$
$$P_{max} = V_{mp}I_{mp} = 249.6 \text{ W}$$
$$\eta = \frac{P_{max}}{A \times G} = \frac{249.6}{1.6 \times 1000} = 0.156 = 15.6\%$$

Q2: Module has $V_{oc}=38V$ at 25°C with temperature coefficient $-0.33\%/^\circ\text{C}$. Calculate V_{oc} at 65°C .

Solution:

$$\Delta T = 65 - 25 = 40^\circ\text{C}$$
$$\Delta V = 38 \times (-0.0033 \times 40) = 38 \times (-0.132) = -5.02 \text{ V}$$
$$V_{oc,65} = 38 - 5.02 = 32.98 \text{ V}$$

Q3: Design a PV array to provide $600V$, $20A$ using modules with $V_{mp}=30V$, $I_{mp}=8A$.

Solution:

$$\text{Modules in series: } N_s = \frac{600}{30} = 20$$
$$\text{Modules in parallel: } N_p = \frac{20}{8} = 2.5 \rightarrow 3 \text{ (round up)}$$
$$\text{Total modules} = 20 \times 3 = 60$$
$$\text{Actual current} = 3 \times 8 = 24 \text{ A } (>20\text{A requirement})$$

Q4: A 5 kW system has performance ratio 0.78 . Location receives $1800 \text{ kWh/m}^2/\text{year}$ irradiation. Calculate annual energy production.

Solution:

$$E = P_{rated} \times G \times PR = 5 \times 1800 \times 0.78 = 7020 \text{ kWh/year}$$

Q5: Using Fractional Open Circuit Voltage method with $k=0.78$, module $V_{oc}=45V$. Find approximate MPP voltage.

Solution:

$$V_{mp} = k \times V_{oc} = 0.78 \times 45 = 35.1 V$$

Q6: In a string of 10 series modules, one module is shaded and produces 50% current. Calculate power loss percentage.

Solution:

Entire string current limited to 50% → Power = 50% of unshaded
Power loss = 50%

Q7: A system produces 20 kWh/day, needs 3 days autonomy with 50% depth of discharge. Battery voltage is 48V. Calculate required Ah capacity.

Solution:

Total storage needed = $20 \times 3 = 60 kWh$
Considering DOD: $Battery\ capacity = \frac{60}{0.5} = 120 kWh$
Ah capacity = $\frac{120 \times 1000}{48} = 2500 Ah$

Q8: A 5 kW inverter has 97% efficiency at 80% load. For 3 kW output, calculate input DC power.

Solution:

$$P_{in} = \frac{P_{out}}{\eta} = \frac{3000}{0.97} = 3093 W$$

Q9: Module temperature is 48°C at 800 W/m² irradiance, ambient 20°C, wind speed 1 m/s. Calculate NOCT.

Solution:

$$\begin{aligned} NOCT &= T_{amb} + \frac{T_{cell} - T_{amb}}{G} \times G_{NOCT} \\ &= 20 + \frac{48 - 20}{800} \times 800 = 20 + 28 = 48^\circ C \end{aligned}$$

Q10: A 10 kW PV system costs ₹5,00,000, produces 15,000 kWh/year. Electricity cost is ₹8/kWh. Calculate simple payback period.

Solution:

$$\text{Annual savings} = 15000 \times 8 = ₹1,20,000$$

$$\text{Payback} = \frac{500000}{120000} = 4.17 \text{ years}$$

MODULE 5: GRID INTEGRATION ISSUES

5.1 Grid Code Requirements

International Grid Code Evolution:

Germany (E.ON Netz): Pioneer in wind integration requirements (2003)

Spain (PO 12.3): Comprehensive technical requirements

USA (FERC 661-A, 827): Federal Energy Regulatory Commission standards

India (CEA Grid Code): Revised 2019 for renewable integration

Technical Requirements:

Voltage Regulation:

Range: Typically $\pm 5\%$ to $\pm 10\%$ of nominal

Ride-through: Must remain connected during specified dips

Reactive Power Capability: Usually ± 0.95 power factor range

Frequency Requirements:

Operating Range: 49.5-50.5 Hz (India), 59.3-60.5 Hz (USA)

Ride-through: Must tolerate specified frequency deviations

Frequency Response:

Inertial response (synthetic inertia)

Primary frequency control (droop response)

Secondary frequency control (AGC participation)

Power Quality Standards:

Harmonics: IEEE 519 limits (THD < 5%, individual harmonics < 3%)

Flicker: IEC 61400-21 measurement and limits

Voltage Unbalance: Typically < 2%

DC Injection: < 0.5% of rated current

5.2 Fault Ride-Through (FRT) Capability

Low Voltage Ride-Through (LVRT):

Purpose: Prevent massive generation loss during grid faults

Requirements:

Must remain connected for specified voltage dips

Must inject reactive current to support voltage recovery

Typical curve: 0% voltage for 150 ms, then recovery to 90% within 1.5 s

Voltage-Dip Categories:

Shallow Dips (>70%): No disconnection allowed

Moderate Dips (50-70%): Ride-through for 500-1000 ms

Deep Dips (15-50%): Ride-through for 250-500 ms

Very Deep Dips (<15%): May disconnect after 150 ms

Reactive Current Injection:

Typically required during voltage dips < 90%

Proportional to voltage deviation: $I_q = k(V_{ref} - V)$

Minimum requirement: 1-2% reactive current per 1% voltage dip

Implementation Techniques:**DFIG LVRT:****Crowbar Protection:**

Shorts rotor windings via resistors

Protects rotor-side converter from overcurrent

Disables converter control temporarily

Modern: Fast crowbar with demagnetization control

Demagnetization Control:

Rapid reduction of rotor flux
Prevents DC link overvoltage
Enables faster recovery

Braking Chopper:

DC link overvoltage protection
Dissipates excess energy in resistor

PMSG LVRT:

DC Chopper: Limits DC link voltage

Current Limiting: Converter current control

Grid Support: Controlled reactive injection

5.3 Active Power Control

Power Ramping Limits:

Up-ramp: Typically 10-20% of rated power per minute

Down-ramp: Typically 10-20% of rated power per minute

Purpose: Prevent sudden changes affecting grid stability

Frequency Response Capabilities:**Inertial Response (Synthetic Inertia):**

Principle: Release stored kinetic energy during frequency drops

DFIG Implementation:

Temporary increase in electrical power

Controlled deceleration of rotor

Power proportional to rate of change of frequency (ROCOF)

Primary Frequency Control:**Droop Characteristic:**

$$\Delta P = -1/R \Delta f$$

Where R = droop constant (3-5% typically)

Deadband: ± 0.01 - 0.03 Hz (no response within deadband)

Response Time: < 2-10 seconds

Secondary Frequency Control (AGC):

Restoration: Returns frequency to nominal

Participation: Renewable plants may provide regulation service

Market Participation: Through ancillary service markets

Delta Control:

Purpose: Prevent overgeneration at high frequency

Implementation: Power reduction above frequency threshold

5.4 Reactive Power and Voltage Control

Reactive Power Capability:

Range: Typically ± 0.95 power factor

At Partial Load: Full reactive capability maintained

Control Modes:

Power Factor Control: Fixed or variable

Voltage Control: Q-V droop characteristic

Reactive Power Control: Setpoint based

Voltage Regulation Strategies:

On-load Tap Changer (OLTC): Transformer voltage adjustment

Static VAR Compensator (SVC): Thyristor-controlled reactors/capacitors

STATCOM: Voltage source converter-based

Wind Plant Capabilities: Modern turbines provide voltage control

Q-V Droop Characteristic:

$$Q = Q_{\text{ref}} + K(V_{\text{ref}} - V)$$

Where K = droop gain (typically 2-10%)

5.5 Power Quality Issues

Voltage Fluctuations and Flicker:

Causes:

Wind turbulence (3P effect from tower shadow)

Cloud transients for solar

Switching operations

Measurement: P_{st} (short-term), P_{lt} (long-term)

Mitigation:

Energy storage (batteries, flywheels)

STATCOM/SVC

Pitch control smoothing

Harmonics:

Sources: Power electronic converters (6-pulse, 12-pulse)

Characteristic Harmonics: $h=6k\pm 1$ for 6-pulse

Mitigation:

Multi-pulse converters (12, 18, 24-pulse)

Active filters

PWM with high switching frequency

Interharmonics and Subsynchronous Oscillations:

Causes: Converter switching, control interactions

Concerns: Resonance with grid components

Monitoring: Wide-area measurement systems

5.6 International Integration Experiences

Germany (Energiewende Experience):

Penetration Level: >50% renewable at times

Challenges:

Grid congestion (North-South transmission)

Frequency stability with reduced inertia

Renewable curtailment during surplus

Solutions:

Grid expansion (HVDC corridors)

Market redesign (balancing markets)

Demand response integration

Denmark (World Leader):

Penetration: >50% annual wind generation

Success Factors:

Strong interconnections (Nordic, German grids)

Flexible generation (CHP plants)

Market integration (Nord Pool)

USA (ERCOT Texas Experience):

Wind Capacity: >30 GW in Texas

Challenges:

Congestion in West Texas

Frequency response with high wind

Hurricane resilience

Solutions:

CREZ transmission lines

Fast frequency response requirements

Energy storage mandate

India (Green Energy Corridors):

RE-Rich States: Tamil Nadu, Gujarat, Rajasthan, Karnataka

Transmission Projects:

ISTS (Inter-State Transmission System)

Green Energy Corridors Phase I & II

Challenges:

Must-run status conflicts

Grid stability with weak interconnections

Forecasting and scheduling compliance

5.7 Hybrid and Isolated Systems

Wind-Solar Hybrid Systems:

Complementarity:

Temporal: Wind higher at night, solar during day

Seasonal: Wind higher in monsoon, solar in summer

Geographical optimization

Benefits:

Smoother power output

Better capacity utilization

Reduced storage requirements

Improved reliability

Battery Energy Storage Integration:

Applications:

Ramp Rate Control: Smoothing power variations

Frequency Regulation: Fast response service

Energy Time-shift: Store excess, discharge during peak

Black Start: System restoration capability

Technologies:

Lithium-ion (dominant)

Flow batteries (long duration)

Advanced lead-acid

Emerging technologies (sodium-ion, solid-state)

Diesel Generator Integration:

Role: Backup for isolated systems

Operational Modes:

Continuous operation (base load)

Peak shaving

Backup/standby

Hybrid Optimization: Minimize fuel consumption through optimal dispatch

Microgrids and Islanded Operation:

Control Challenges:

Frequency stability without grid connection

Voltage regulation with variable generation

Load-generation balance

Control Strategies:

Master-slave control

Peer-to-peer control

Droop control methods

5.8 AI Applications in Grid Integration

Forecasting and Scheduling:

Very Short-term (5 min-1 hour): ANN, LSTM for cloud movement

Day-ahead: Ensemble methods combining NWP with ML

Renewable Generation Prediction:

Wind: Numerical weather models + ML correction

Solar: Sky imaging + satellite data + ML

Optimal Dispatch and Unit Commitment:

Stochastic Optimization: Considering forecast uncertainty

Machine Learning: For fast approximation of complex problems

Reinforcement Learning: Adaptive control strategies

Anomaly Detection and Fault Diagnosis:

PMU Data Analysis: Deep learning for grid disturbance classification

Protection Scheme Optimization: Adaptive relay settings

Cyber-security: ML for intrusion detection in SCADA systems

Grid Stability Assessment:

Real-time Stability Margins: ML for fast computation

Voltage Stability Prediction: Pattern recognition from PMU data

Oscillation Detection: Spectral analysis with ML classification

:IMPORTANT QUESTION:

Q1: What are the key grid code requirements for renewable integration?

A1:

1. **Voltage regulation:** $\pm 5\%$ -10% of nominal
2. **Frequency range:** 49.5-50.5 Hz (India)
3. **Power factor:** 0.95 lagging to 0.95 leading
4. **Fault ride-through:** LVRT/HVRT capability
5. **Harmonics:** THD <5%, individual <3% (IEEE 519)
6. **Ramp rates:** Limited power change rates

Q2: Explain LVRT requirements and implementation in DFIG.

A2: LVRT requires staying connected during voltage dips:

Requirements: Typically 0% voltage for 150ms, recovery to 90% in 1.5s

DFIG Implementation:

- Crowbar activation to protect converter
- Reactive current injection (1-2% per 1% voltage dip)
- Demagnetization control for flux reduction
- DC chopper for overvoltage protection

Q3: What are power quality issues with wind integration?

A3:

Voltage fluctuations/flicker: From wind turbulence

Harmonics: From power converters

Voltage unbalance: From uneven generation

Frequency variations: From intermittency

Reactive power demand: From induction generators

Q4: How does AI help in grid integration?

A4:

Forecasting: ML for generation prediction

Optimal dispatch: Optimization algorithms

Fault detection: Pattern recognition

Stability assessment: Real-time analysis

Protection coordination: Adaptive settings

: NUMERICAL PROBLEMS:

Q1: A 100 MW wind farm must operate at ± 0.95 power factor. Calculate reactive power range.

Solution:

$$\theta = \cos^{-1}(0.95) = 18.19^\circ$$

$$Q = P \times \tan \theta = 100 \times \tan(18.19^\circ) = 100 \times 0.3287 = \pm 32.87 \text{ MVAr}$$

Q2: During voltage dip to 70%, reactive current injection required is 2% of rated current per 1% voltage dip. System rated 100 MVA, 0.8 PF at 33 kV. Calculate required reactive current.

Solution:

$$\text{Voltage dip} = 30\%$$

$$\text{Required reactive current} = 30 \times 2 = 60\% \text{ of rated current}$$

$$\text{Rated current} = \frac{100 \times 10^6}{\sqrt{3} \times 33 \times 10^3} = 1749.5 \text{ A}$$

$$\text{Reactive current} = 0.6 \times 1749.5 = 1049.7 \text{ A}$$

Q3: Design capacitor for 5th harmonic filter. 5th harmonic current is 50A at 33kV, 50Hz fundamental.

Solution:

$$\text{For 5th harmonic: } f = 250 \text{ Hz}$$

$$X_c = \frac{V}{I} = \frac{33000/\sqrt{3}}{50} = \frac{19052.6}{50} = 381 \Omega$$

$$C = \frac{1}{2\pi f X_c} = \frac{1}{2\pi \times 250 \times 381} = 1.67 \mu\text{F}$$

Q4: Wind farm has 5% droop. Frequency drops from 50 Hz to 49.5 Hz. Calculate percentage power increase.

Solution:

$$\Delta f = -0.5 \text{ Hz}$$
$$\Delta P = -\frac{1}{R} \times \frac{\Delta f}{f_{rated}} \times 100\% = -\frac{1}{0.05} \times \frac{-0.5}{50} \times 100 = 20\% \text{ increase}$$

Q5: A 200 MW wind farm has ramp rate limit of 10%/minute. What is maximum power change in 5 minutes?

Solution:

$$\text{Maximum change per minute} = 200 \times 0.10 = 20 \text{ MW}$$
$$\text{In 5 minutes} = 20 \times 5 = 100 \text{ MW}$$

Q6: To smooth ± 20 MW fluctuations from 50 MW wind farm over 15 minutes, calculate battery energy storage requirement.

Solution:

$$\text{Time} = 15 \text{ minutes} = 0.25 \text{ hours}$$
$$\text{Energy} = \text{Power} \times \text{time} = 20 \times 0.25 = 5 \text{ MWh}$$

Q7: A 50 MW wind farm operates at 0.8 lagging PF. Calculate capacitor bank rating to improve PF to 0.95 lagging.

Solution:

$$\text{At 0.8 PF: } Q_1 = 50 \times \tan(\cos^{-1} 0.8) = 50 \times 0.75 = 37.5 \text{ MVAr}$$
$$\text{At 0.95 PF: } Q_2 = 50 \times \tan(\cos^{-1} 0.95) = 50 \times 0.3287 = 16.44 \text{ MVAr}$$
$$\text{Capacitor rating} = Q_1 - Q_2 = 37.5 - 16.44 = 21.06 \text{ MVAr}$$

Q8: A 60 MVA transformer serves 45 MW wind farm at 0.95 PF. Calculate transformer loading percentage.

Solution:

$$\text{Apparent power} = \frac{P}{PF} = \frac{45}{0.95} = 47.37 \text{ MVA}$$

$$\text{Loading} = \frac{47.37}{60} \times 100 = 78.95\%$$

Q9: 20 MW injection at 33kV causes voltage rise. Grid has impedance $0.2+j0.5 \Omega$. Calculate voltage rise magnitude.

Solution:

$$\text{Current} = \frac{20 \times 10^6}{\sqrt{3} \times 33000} = 349.9 \text{ A}$$

$$\text{Voltage rise} = I \times R = 349.9 \times 0.2 = 70 \text{ V}$$

$$\text{Percentage rise} = \frac{70}{33000/\sqrt{3}} \times 100 = \frac{70}{19052.6} \times 100 = 0.37\%$$

Q10: A 100 MW wind farm is curtailed 20% for 200 hours annually. Calculate energy loss and revenue loss at ₹4/kWh.

Solution:

$$\text{Energy loss} = 100 \times 0.20 \times 200 = 4000 \text{ MWh} = 4,000,000 \text{ kWh}$$

$$\text{Revenue loss} = 4,000,000 \times 4 = ₹16,000,000 = ₹1.6 \text{ crores}$$

MODULE 6: SOLAR THERMAL POWER GENERATION

6.1 Concentrating Solar Power (CSP) Technologies

Parabolic Trough Systems:

Design: Parabolic curved mirrors focusing sunlight onto receiver tube

Tracking: Single-axis (N-S horizontal or polar aligned)

Working Fluid: Thermal oil (up to 400°C) or direct steam

Concentration Ratio: 70-80 suns

Typical Capacity: 50-280 MW per plant

Efficiency: 14-20% (solar to electric)

Storage Integration: 2-tank molten salt storage (up to 7.5 hours)

Major Installations:

SEGS plants, California (354 MW total)

Andasol 1-3, Spain (150 MW with 7.5h storage)

NOORo complex, Morocco (510 MW with storage)

Central Receiver (Power Tower) Systems:

Design: Field of heliostats focusing on central receiver atop tower

Tracking: Dual-axis for each heliostat

Working Fluid: Molten salt (290-565°C), steam, air

Concentration Ratio: 800-1000 suns

Temperature: 500-1000°C

Efficiency: 20-35% (higher operating temperature)

Advantages:

Higher efficiency and temperature

Better storage integration

Reduced piping losses

Installations:

Ivanpah, USA (377 MW, dry cooling)

Cerro Dominador, Chile (110 MW with 17.5h storage)

Crescent Dunes, USA (110 MW with 10h storage)

Parabolic Dish Systems:

Design: Parabolic dish with receiver at focal point

Tracking: Dual-axis

Power Conversion: Stirling engine or microturbine

Concentration Ratio: 1000-3000 suns

Temperature: 650-750°C

Efficiency: 25-30% (highest among CSP)

Advantages:

Modular design

No water requirement

High efficiency

Disadvantages:

High cost per unit
Complex tracking

Installations:

Maricopa Solar, USA (1.5 MW, Stirling engines)
Tooele Army Depot, USA

Linear Fresnel Reflector (LFR) Systems:

Design: Multiple flat or slightly curved mirrors focusing on linear receiver

Tracking: Single-axis for each mirror row

Receiver: Fixed elevated tube

Concentration Ratio: 25-100 suns

Temperature: 270-550°C

Efficiency: 8-12%

Advantages:

- Lower mirror cost (flat/less curved)
- Fixed receiver (simplifies piping)
- Ground coverage optimization

Disadvantages:

- Lower efficiency
- Optical losses

Installations:

- Puerto Errado, Spain (30 MW)
- Dhursar, India (125 MW)

6.2 Solar Pond Technology

Principles and Design:

Non-convecting Principle: Salt gradient prevents natural convection

Three-layer Structure:

Upper Convective Zone (UCZ): Low salinity, 0.5-1m thick

Non-convective Zone (NCZ): Salinity gradient, 1-1.5m thick

Lower Convective Zone (LCZ): High salinity, heat storage, 1.5-2m thick

Construction and Operation:

Pond Dimensions: Typically 0.5-5m depth, area varies (100 m² to km²)

Salt Types: NaCl, MgCl₂ (higher solubility preferred)

Temperature Gradient: Surface: ambient, Bottom: 80-90°C

Heat Extraction: Heat exchanger in LCZ

Applications:

1. Industrial process heat
2. Power generation (ORC cycle)
3. Desalination
4. Space heating

Advantages and Limitations:**Advantages:**

- Large heat storage capacity (seasonal storage possible)
- Low construction cost
- Simple technology

Disadvantages:

- Large land requirement
- Salt management
- Maintenance challenges
- Low power cycle efficiency

Performance Parameters:

1. **Efficiency:** 15-25% (solar to thermal)
2. **Heat Extraction Efficiency:** 20-30%
3. **Temperature Stability:** ±5°C with proper management

6.3 Thermal Performance Analysis

Optical Efficiency (η_{opt}):

$$\eta_{opt} = \rho\tau\alpha\gamma \cos \theta$$

Where:

ρ = mirror reflectivity (0.90-0.95)

τ = receiver cover transmissivity (0.90-0.95)

α = receiver absorptivity (0.90-0.95)

γ = intercept factor (0.95-0.98)

$\cos \theta$ = cosine loss (tracking error)

Thermal Efficiency (η_{th}):

$$\eta_{th} = \frac{Q_u}{Q_{abs}} = 1 - \frac{U_L(T_r - T_a)}{C \cdot G_b}$$

Where:

Q_u = useful heat gain

Q_{abs} = absorbed solar radiation

U_L = overall heat loss coefficient (W/m²K)

T_r = receiver temperature

T_a = ambient temperature

C = concentration ratio

G_b = beam radiation

Overall System Efficiency:

$$\eta_{system} = \eta_{opt} \times \eta_{th} \times \eta_{power\ cycle}$$

Where $\eta_{power\ cycle}$ = 35-45% (Rankine cycle efficiency)

Capacity Factor Enhancement:

- **Without Storage:** 25-35%
- **With Storage (6-15 hours):** 40-60%
- **Hybridization (with natural gas):** 70-85%

6.4 Heat Transfer Fluids and Storage

Working Fluids:**Thermal Oils:**

Synthetic (e.g., Therminol VP-1)

Temperature limit: ~400°C

Disadvantages: Flammable, environmental concerns

Molten Salts:

Binary (NaNO₃-KNO₃, 60-40%)

Temperature range: 290-565°C

Advantages: Good thermal properties, low vapor pressure

Disadvantages: Freezing point high (220°C), corrosion issues

Water/Steam:

Direct steam generation (DSG)

Advantages: No HTF cost, higher temperature

Disadvantages: Two-phase flow complexity, pressure control

Liquid Metals:

Sodium, lead-bismuth

Advantages: Excellent heat transfer, high temperature

Disadvantages: Safety concerns, corrosion

Air/Other Gases:

Helium, CO₂ (supercritical)

Advantages: High temperature, no freezing

Disadvantages: Low density, poor heat transfer

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Thermal Energy Storage (TES):**Sensible Heat Storage:**

Two-tank system: Hot and cold salt tanks

Single-tank thermocline: Temperature gradient in single tank

Storage medium: Molten salt, concrete, rocks

Latent Heat Storage:

Phase change materials (PCM)

Higher energy density

Isothermal storage

Materials: Salt hydrates, metals, paraffin

Thermochemical Storage:

Reversible chemical reactions

Highest energy density

Long-duration storage potential

Examples: Metal hydrides, ammonia dissociation

Storage Duration Economics:

Short-term (1-6 hours): For daily cycling

Medium-term (6-15 hours): For evening peak and cloudy periods

Long-term (> 15 hours): For multiple cloudy days, seasonal storage

6.5 Power Cycle Integration

Steam Rankine Cycle:

- **Configuration:** Subcritical or supercritical
- **Turbine Inlet:** 370-550°C, 100-160 bar
- **Condensing:** Water-cooled or air-cooled
- **Efficiency:** 35-42% (solar to electric)

Organic Rankine Cycle (ORC):

- **Working Fluid:** Organic compounds (pentane, toluene)
- **Applications:** Lower temperature sources (<300°C)
- **Efficiency:** 15-25%
- **Advantages:** Simpler operation, dry cooling compatible

Brayton Cycle (Gas Turbine):

- **Integration:** Solar-hybrid gas turbines
- **Temperature:** 800-1000°C
- **Efficiency:** 30-40%
- **Advantages:** Dry cooling, fast startup

Combined Cycle:

- **Configuration:** Brayton + Rankine bottoming cycle
- **Efficiency:** 45-50%
- **Applications:** ISCC (Integrated Solar Combined Cycle)

6.6 Economic and Environmental Aspects

Cost Analysis:

Capital Cost: \$4,000-9,000/kW (higher than PV)

LCOE: \$0.12-0.25/kWh (dependent on DNI and storage)

Cost Drivers:

- Solar field (50-60% of total)
- Power block (20-25%)
- Storage system (10-20%)

Learning Curve: 10-15% cost reduction per doubling of capacity

Environmental Impacts:

Land Use: 4-6 acres/MW (higher than PV)

Water Consumption:

Wet cooling: 3,000-4,000 L/MWh

Dry cooling: 300-400 L/MWh (mostly for mirror cleaning)

Lifecycle Emissions: 15-25 gCO₂eq/kWh (vs. 800-1000 for coal)

Comparative Analysis with PV:

Parameter	CSP with Storage	Utility-scale PV + Battery
Dispatchability	Inherent with TES	Limited by battery size
Capacity Factor	40-60%	15-25% (PV only)
Land Use	Higher	Lower
Water Use	Higher (unless dry cooled)	Minimal
Technology Maturity	Moderate	High
Cost Trend	Decreasing slowly	Rapidly decreasing

INTEGRATED THEMES AND EMERGING TRENDS

Hybrid Renewable Systems

Wind-Solar Complementary Analysis:

Temporal: Correlation coefficients analysis

Geographical: Resource mapping optimization

System Sizing: Multi-objective optimization

Artificial Intelligence Integration

Across Technology Stack:

Resource Assessment: ML for site selection

Design Optimization: AI for component sizing

Operational Control: RL for adaptive control

Maintenance: Predictive analytics

Grid Integration: AI for forecasting and dispatch

Specific Applications:

Digital Twins: Virtual replicas for simulation and optimization

Transfer Learning: Knowledge transfer between sites

Explainable AI (XAI): For regulatory compliance and trust

Policy and Regulatory Framework

International Best Practices:

Feed-in Tariffs: Germany, Spain (early adoption)

Renewable Portfolio Standards: USA, India

Auction Mechanisms: Competitive bidding (most countries)

Green Certificates: Tradable renewable certificates

Net Metering: Rooftop solar promotion

Grid Integration Policies:

Forecasting and Scheduling: Penalties for deviation

Ancillary Services Markets: Participation of renewables

Grid Code Evolution: Dynamic requirements based on penetration

Future Research Directions

Materials Science:

Perovskite stability improvement

Alternative rare-earth materials

High-temperature materials for CSP

System Integration:

Hydrogen production integration

Multi-energy systems (electricity, heat, transport)

Virtual power plants

Digitalization:

Blockchain for peer-to-peer trading

IoT for asset management

Cybersecurity for smart grids

Circular Economy:

PV panel recycling
Wind turbine blade recycling
Sustainable material sourcing

:IMPORTANT QUESTION:

6.1 CSP Technologies and Performance

Q1: Compare different CSP technologies.

A1:

Technology	Focus	Temp (°C)	Efficiency	Applications
Parabolic trough	Linear	300-400	14-20%	Utility scale
Power tower	Point	500-1000	20-35%	High temp apps
Parabolic dish	Point	650-750	25-30%	Distributed
Linear Fresnel	Linear	270-550	8-12%	Low cost

Q2: Explain solar pond working principle.

A2: Solar pond uses salt gradient to prevent convection:

Upper Convective Zone: Low salinity, mixes

Non-convective Zone: Salinity gradient, inhibits convection

Lower Convective Zone: High salinity, stores heat (80-90°C)

Heat extracted from bottom for power generation (ORC) or process heat.

Q3: What are the efficiency components in CSP systems?

A3:

Optical efficiency (η_{opt}): Mirror reflectivity, intercept factor, cosine losses

Thermal efficiency (η_{th}): Receiver absorption, heat losses

Power cycle efficiency (η_{cycle}): Rankine cycle efficiency

Overall: $\eta_{\text{system}} = \eta_{\text{opt}} \times \eta_{\text{th}} \times \eta_{\text{cycle}}$

Q4: Describe thermal energy storage methods in CSP.

A4:

Sensible heat: Molten salt (most common), 2-tank or thermocline

Latent heat: Phase change materials, higher density

Thermochemical: Reversible reactions, highest density

Solid media: Concrete, rocks

:NUMERICAL PROBLEMS:

Q1: A 50 MW parabolic trough plant has 20% efficiency. DNI is 800 W/m². Calculate required collector area.

Solution:

$$\begin{aligned}\text{Solar power needed} &= \frac{50}{0.20} = 250 \text{ MW} \\ \text{Area} &= \frac{250 \times 10^6}{800} = 312,500 \text{ m}^2 = 31.25 \text{ hectares}\end{aligned}$$

Q2: Parabolic trough has aperture width 5m, receiver diameter 70mm. Calculate geometric concentration ratio.

Solution:

$$C = \frac{\text{Aperture}}{\text{Receiver diameter}} = \frac{5}{0.07} = 71.4$$

Q3: Receiver temperature 350°C, ambient 25°C, heat loss coefficient 15 W/m²K, concentration ratio 80, DNI 900 W/m². Calculate thermal efficiency.

Solution:

$$\begin{aligned}\eta_{th} &= 1 - \frac{U_L(T_r - T_a)}{C \times G_b} = 1 - \frac{15 \times (350 - 25)}{80 \times 900} \\ &= 1 - \frac{15 \times 325}{72,000} = 1 - 0.0677 = 0.932 = 93.2\%\end{aligned}$$

Q4: A 100 MW CSP plant needs 6 hours storage. Thermal efficiency 90%, power cycle efficiency 38%. Molten salt has Cp=1.5 kJ/kgK, ΔT=300°C. Calculate salt mass required.

Solution:

$$\begin{aligned}\text{Thermal energy needed} &= \frac{100 \times 6}{0.38} = 1578.9 \text{ MWh} \\ \text{In joules} &= 1578.9 \times 3.6 \times 10^9 = 5.684 \times 10^{12} \text{ J} \\ \text{Mass} &= \frac{E}{C_p \Delta T} = \frac{5.684 \times 10^{12}}{1500 \times 300} = 12.63 \times 10^6 \text{ kg} = 12,630 \text{ tons}\end{aligned}$$

Q5: Solar pond area 50,000 m² receives average insolation 200 W/m². Heat extraction is 3 MW. Calculate pond efficiency.

Solution:

$$\text{Solar input} = 50,000 \times 200 = 10,000,000 \text{ W} = 10 \text{ MW}$$

$$\text{Efficiency} = \frac{3}{10} = 0.30 = 30\%$$

Q6: A 100 MW CSP plant with 25% efficiency, DNI 2500 kWh/m²/year, land factor 4. Calculate land area required.

Solution:

Annual energy Area

$$\text{Annual energy} = 100 \times 8760 \times 0.25 = 219,000 \text{ MWh (assuming 25\% CF)}$$

$$\text{Solar energy needed} = \frac{219,000}{0.25} = 876,000 \text{ MWh}$$

$$\text{Area} = \frac{876,000 \times 1000}{2500} = 350,400 \text{ m}^2 = 35.04 \text{ hectares}$$

$$\text{With land factor: } 35.04 \times 4 = 140.16 \text{ hectares}$$

Q7: Power tower with heliostat field radius 500m needs 1° tracking accuracy. Estimate tower height for minimum cosine loss.

Solution:

For minimum cosine loss: $H \approx R = 500 \text{ m}$

Q8: 100 MW plant with wet cooling uses 3.5 m³/MWh. Capacity factor 25%. Calculate annual water consumption.

Solution:

$$\text{Annual generation} = 100 \times 8760 \times 0.25 = 219,000 \text{ MWh}$$

$$\text{Water consumption} = 219,000 \times 3.5 = 766,500 \text{ m}^3$$

Q9: Plant cost \$4000/kW for 100 MW, O&M 2% of capital annually, capacity factor 25%, lifetime 25 years. Estimate LCOE.

Solution:

$$\text{Capital cost} = 100,000 \times 4000 = \$400 \text{ million}$$

$$\text{Annual O\&M} = 0.02 \times 400 = \$8 \text{ million}$$

$$\text{Annual generation} = 100,000 \times 8760 \times 0.25 = 219 \text{ million kWh}$$

$$\text{LCOE} \approx \frac{400/25 + 8}{219} = \frac{16 + 8}{219} = \$0.1096/\text{kWh}$$

Q10: Hybrid plant gets 60% heat from solar (20% efficiency) and 40% from natural gas (45% efficiency). Calculate overall efficiency.

Solution:

$$\text{Overall efficiency} = 0.6 \times 0.20 + 0.4 \times 0.45 = 0.12 + 0.18 = 0.30 = 30\%$$

:NUMERICAL PROBLEMS:

Q1: Hybrid wind-solar system:

Wind: 20 MW, CF=32%

Solar: 10 MW, CF=20%

Load: 8 MW constant

Calculate: a) Total annual energy b) Percentage of load met

Solution:

$$\text{a) Wind energy} = 20 \times 0.32 \times 8760 = 56,064 \text{ MWh}$$

$$\text{Solar energy} = 10 \times 0.20 \times 8760 = 17,520 \text{ MWh}$$

$$\text{Total} = 73,584 \text{ MWh}$$

$$\text{b) Load energy} = 8 \times 8760 = 70,080 \text{ MWh}$$

$$\text{Percentage met} = \frac{73,584}{70,080} \times 100 = 105\%$$

Q2: Battery sizing for renewable smoothing:

System has 50 MW wind, 30 MW solar.

Max ramp: 20 MW in 10 minutes

Calculate battery power and energy for smoothing.

Solution:

Power rating = 20 MW

$$\text{Energy} = 20 \times \frac{10}{60} = 3.33 \text{ MWh}$$
